

9

Transport properties in gases

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In this chapter, we wish to describe how a gas can transport momentum, energy or particles, from one place to another. The model we have used so far has been that of a gas in equilibrium, so that none of its parameters are time-dependent. Now we consider *non-equilibrium* situations, but still in the *steady state*, i.e. so that the system parameters are time-independent, but the surroundings will be time-dependent. The phenomena we want to treat are called **transport properties** and we will consider

- (1) **viscosity**, which is the transport of *momentum*,
- (2) **thermal conductivity** which is the transport of *heat*, and
- (3) **diffusion**, which is the transport of *particles*.

9.1 Viscosity

¹This proportionality was suggested by Isaac Newton and holds for many liquids and most gases, which are thus termed **Newtonian fluids**. Non-Newtonian fluids have a viscosity which is a function of the applied shear stress.

²Also used is the **kinematic viscosity** ν , defined by $\nu = \eta/\rho$ where ρ is the density. This is useful because one often wants to compare the viscous forces with inertial forces. The unit of kinematic viscosity is $\text{m}^2 \text{s}^{-1}$.

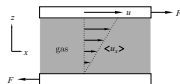


Fig. 9.1 A fluid is sandwiched between two plates of area A which each lie in an xy plane.

Viscosity is the measure of the resistance of a fluid to the deformation produced by a shear stress. For straight, parallel and uniform flow, the shear stress between the layers is proportional¹ to the velocity gradient in the direction perpendicular to the layers. The constant of proportionality, given the symbol η , is called the **coefficient of viscosity**, the **dynamic viscosity** or simply the **viscosity**.²

Consider the scenario in Fig. 9.1 in which a fluid is sandwiched between two plates of area A which each lie in the xy plane. A shear stress $\tau_{xz} = F/A$ is applied to the fluid by sliding the top plate over it at speed u while keeping the bottom plate stationary. A shear force F is applied. A velocity gradient $d\langle u_x \rangle/dz$ is set up, so that $\langle u_x \rangle = 0$ near the bottom plate and $\langle u_x \rangle = u$ near the top plate. If the fluid is a gas, then this extra motion in the x -direction is superimposed on the Maxwell-Boltzmann motion in the x , y and z directions (and hence the use of the average $\langle u_x \rangle$, rather than u_x).

The viscosity η is then defined by

$$\tau_{xz} = \frac{F}{A} = \eta \frac{d\langle u_x \rangle}{dz}. \quad (9.1)$$

The units of viscosity are Pa s ($= \text{N m}^{-2} \text{s}$). Force is rate of change of momentum, and hence transverse momentum is being transported

through the fluid. This is achieved because molecules travelling in the $+z$ direction move from a layer in which $\langle u_x \rangle$ is smaller to one in which $\langle u_x \rangle$ is larger, and hence they transfer net momentum to that layer in the $-x$ direction. Molecules travelling parallel to $-z$ have the opposite effect. Hence, the shear stress τ_{xz} is equal to the transverse momentum transported across each square metre per second, and hence τ_{xz} is equal to a flux of momentum (though note that there must be a minus sign involved, because the momentum flux must be from regions of high transverse velocity to regions of low transverse velocity, which is in the opposite direction to the velocity gradient). The velocity gradient $\partial \langle u_x \rangle / \partial z$ therefore drives a momentum flux Π_z , according to

$$\Pi_z = -\eta \frac{\partial \langle u_x \rangle}{\partial z}. \quad (9.2)$$

The viscosity can be calculated using kinetic theory as follows:

Recall first that we showed before in eqn 6.13 that the number of molecules which hit unit area per second is $v \cos \theta n f(v) dv \frac{1}{2} \sin \theta d\theta$. Consider molecules travelling at an angle θ to the z -axis (see Fig. 9.2). Then molecules which cross a plane of constant z will have travelled on average a distance λ since their last collision, and so they will have travelled a distance $\lambda \cos \theta$ parallel to the z -axis since their last collision. Over that distance there is an average increase in $\langle u_x \rangle$ given by $(\partial \langle u_x \rangle / \partial z) \lambda \cos \theta$, so these upward travelling molecules bring an excess momentum in the x -direction given by³

$$-m \left(\frac{\partial \langle u_x \rangle}{\partial z} \right) \lambda \cos \theta. \quad (9.3)$$

Hence the total x -momentum transported across unit area perpendicular to z in unit time is the momentum flux Π_z given by

$$\begin{aligned} \Pi_z &= \int_0^\infty \int_0^\pi v \cos \theta n f(v) dv \frac{1}{2} \sin \theta d\theta \cdot m \left(-\frac{\partial \langle u_x \rangle}{\partial z} \right) \lambda \cos \theta \\ &= \frac{1}{2} nm \lambda \int_0^\infty v f(v) dv \left(-\frac{\partial \langle u_x \rangle}{\partial z} \right) \int_0^\pi \cos^2 \theta \sin \theta d\theta \\ &= -\frac{1}{3} nm \lambda \langle v \rangle \left(\frac{\partial \langle u_x \rangle}{\partial z} \right). \end{aligned} \quad (9.4)$$

Hence the viscosity is given by

$$\boxed{\eta = \frac{1}{3} nm \lambda \langle v \rangle}. \quad (9.5)$$

Equation 9.5 has some important consequences.

- η independent of pressure.

Because $\lambda \approx 1/(\sqrt{2}n\sigma) \propto n^{-1}$, the viscosity is independent of n and hence (at constant temperature) it is independent of pressure. This is at first sight a weird result: as you increase the pressure, and hence n , you should be better at transmitting momentum

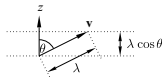


Fig. 9.2 Molecular velocity $\mathbf{v}$ for molecules travelling at an angle θ to the z -axis. These will have travelled on average a distance λ since their last collision, and so they will have travelled a distance $\lambda \cos \theta$ parallel to the z -axis since their last collision.

³The negative sign is because the molecules moving in the $+z$ direction are moving up the velocity gradient from a slower to a faster region and so bring a deficit in x -momentum if $\left(\frac{\partial \langle u_x \rangle}{\partial z} \right)$ is positive. It is the same reason for the negative sign in eqn 9.2.

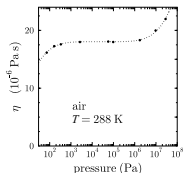


Fig. 9.3 The apparent viscosity of air as a function of pressure at 288 K. It is found to be constant over a wide range of pressure.

because you have more molecules to do it with. However, your mean free path reduces correspondingly, so that each molecule becomes less effective at transmitting momentum in such a way as to precisely cancel out the effect of having more of them.

This result holds impressively well over quite a range of pressures (see Fig. 9.3) although it begins to fail at very low or very high pressures.

- $\eta \propto T^{1/2}$.

Because η is independent of n , the only temperature dependence is from $\langle v \rangle \propto T^{1/2}$, and hence $\eta \propto T^{1/2}$. Note therefore that the viscosity of gases *increases* with T , which is different for most liquids which get runnier (i.e. *less* viscous) when you heat them.

- Substituting in $\lambda = (\sqrt{2}n\sigma)^{-1}$, $\sigma = \pi d^2$ and $\langle v \rangle = (8k_B T / \pi m)^{1/2}$ yields a more useful (though less memorable) expression for the viscosity:

$$\eta = \frac{2}{3\pi d^2} \left(\frac{mk_B T}{\pi} \right)^{1/2}. \quad (9.6)$$

- Equation 9.6 predicts that the viscosity will be proportional to $\sqrt{m}/d^2$ at constant temperature. This holds very well, as shown in Fig. 9.4.

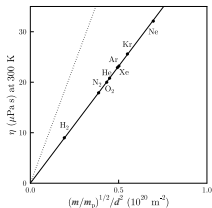


Fig. 9.4 The dependence of the viscosity of various gases on $\sqrt{m}/d^2$. The dotted line is the prediction of eqn 9.6. The solid line is the prediction of eqn 9.45.

- Various *approximations* have gone into this approach, and a condition for their validity is that

$$L \gg \lambda \gg d, \quad (9.7)$$

where L is the size of the container holding the gas and d is the molecular diameter. We need $\lambda \gg d$ (pressure not too high) so that we can neglect collisions involving more than two particles. We need $\lambda \ll L$ (pressure not too low) so that molecules mainly collide with each other and not with the container walls. If λ is of the

same order of magnitude or greater than L , most of a molecule's collisions will be with the container walls. Figure 9.3 indeed shows that the pressure-independence of the viscosity begins to break down when the pressure is too low or too high.

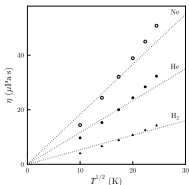


Fig. 9.5 The temperature dependence of the viscosity of various gases. The agreement with the predicted $T^{1/2}$ behaviour is satisfactory as a first approximation, but not very good in detail.

- The factor of $\frac{1}{3}$ in eqn 9.5 is not quite right, so that eqn 9.6 leads to the dotted line in Fig. 9.4. To get a precise numerical factor, you need to consider the fact that the velocity distribution is different in different layers (because of the shear stress applied) and then average over the distribution of path lengths. This will be done in Section 9.4 and leads to a prediction which gives the solid line in Fig. 9.4.
- The measured temperature dependence of the viscosity of various gases broadly agrees with our prediction that $\eta \propto \sqrt{T}$, as shown in Fig. 9.5, but the agreement is not quite perfect. The reason for this is that the collision cross-section, $\sigma = \pi d^2$, is actually temperature-dependent. At high temperatures, molecules move faster and hence have to collide more directly to have a proper momentum-randomizing collision. We have been assuming that molecules behave as perfect hard spheres and that any collision perfectly randomizes the molecular motion, but this is not precisely true. This means that the effective molecular diameter shrinks as you increase the temperature, increasing the viscosity over and above the expected $\sqrt{T}$ dependence. This is evident in the data presented in Fig. 9.5.
- Viscosity can be measured by the damping of torsional oscillations in the apparatus shown in the box.

Measurement of viscosity

Maxwell developed a method for measuring the viscosity of a gas by observing the damping rate of oscillations of a disk suspended from a fixed support by a torsion fibre.

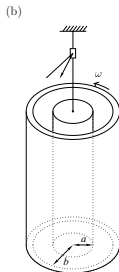
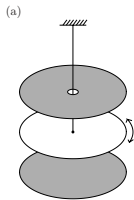


Fig. 9.6 Measuring viscosity by (a) Maxwell's method and (b) the rotating cylinder method.

It is positioned halfway between two, fixed horizontal disks and oscillates parallel to them in the gas. This is shown in Fig. 9.6(a), with the fixed horizontal disks shaded and the oscillating disk in white. The damping of the torsional oscillations is from the viscous damping due to the gas trapped on each side of the oscillating disk between the fixed disks. The fixed disks are mounted inside a vacuum chamber in which the composition and pressure of the gas to be measured can be varied.

A very accurate method is the *rotating-cylinder method* in which gas is confined between two vertical coaxial cylinders. It is shown in Fig. 9.6(b). The outer cylinder (inner radius a) is rotated by a motor at a constant angular speed ω_0 , while the inner cylinder (outer radius b) is suspended by a torsion fibre from a fixed support. The torque G on

the outer cylinder is transmitted via the gas to the inner cylinder and a resulting torque on the torsion

fibre. The velocity gradient $u(r)$ is related to the angular velocity $\omega(r)$ by $u(r) = r\omega(r)$, and we expect that ω varies all the way from 0 at $r = a$ to ω_0 at $r = b$. The velocity gradient is thus

$$\frac{du}{dr} = \omega + r \frac{d\omega}{dr}, \quad (9.8)$$

but the first term here simply corresponds to the velocity gradient due to rigid rotation and does not contribute to the viscous shearing stress which is thus $\eta r d\omega/dr$. The force F on a cylindrical element of gas (of length l) is then just this viscous stress multiplied by the area of the cylinder $2\pi r l$, i.e.

$$F = 2\pi r l \eta \times r \frac{d\omega}{dr}, \quad (9.9)$$

and so the torque $G = rF$ on this cylindrical element is

$$G = 2\pi r^3 l \eta \frac{d\omega}{dr}. \quad (9.10)$$

In the steady state, there is no change in viscous torque from the outer to the inner cylinder (if there were, angular acceleration would be induced somewhere and the system would change) so this torque is transmitted to the suspended cylinder. Hence rearranging and integrating give

$$G \int_a^b \frac{dr}{r^3} = 2\pi l \eta \int_0^{\omega_0} d\omega = 2\pi l \eta \omega_0, \quad (9.11)$$

so that

$$\eta = \frac{G}{4\pi \omega_0 l} \left(\frac{1}{a^2} - \frac{1}{b^2} \right). \quad (9.12)$$

The torque G is related to the angular deflection ϕ of the inner cylinder by $G = \alpha \phi$. The angular deflection can be measured using a light beam reflected from a small mirror attached to the torsion fibre. The coefficient α is known as the **torsion constant**. This can be found by measuring the period T of torsional oscillations of an object of moment of inertia I suspended from the wire, which is

$$T = 2\pi \sqrt{\frac{I}{\alpha}}. \quad (9.13)$$

Knowledge of I and T yields α which can be used with the measured ϕ to obtain G and hence η .

9.2 Thermal conductivity

We have defined **heat** as ‘energy in transit’.⁴ It quantifies the transfer of energy in response to a temperature gradient. The amount of heat which flows along a temperature gradient depends upon the thermal conductivity of the material which we will now define.

Thermal conductivity can be considered in one-dimension using the diagram shown in Fig. 9.7. Heat flows from hot to cold, and so flows against the temperature gradient. The flow of heat can be described by a heat flux vector $\mathbf{J}$, whose direction lies along the direction of flow of heat and whose magnitude is equal to the heat energy flowing per unit time per second (measured in $\text{J s}^{-1} \text{m}^{-2} = \text{W m}^{-2}$). The heat flux J_z in the z -direction is given by

$$J_z = -\kappa \left(\frac{\partial T}{\partial z} \right), \quad (9.14)$$

where the negative sign is because heat flows ‘downhill’. The constant κ is called the **thermal conductivity**⁵ of the gas. In general, in three dimensions we can write that the heat flux $\mathbf{J}$ is related to temperature using

$$\mathbf{J} = -\kappa \nabla T. \quad (9.15)$$

How do molecules in a gas ‘carry’ heat? Gas molecules have energy, and as we found in eqn 5.17 their mean translational kinetic energy ($\frac{1}{2}mv^2$) = $\frac{3}{2}k_B T$ depends on the temperature. Therefore to increase the temperature of a gas by 1 K, one has to increase the mean kinetic energy by $\frac{3}{2}k_B$ per molecule. The heat capacity⁶ C of the gas is the heat required to increase the temperature of gas by 1 K. The heat capacity C_{molecule} of a gas molecule is therefore equal to $\frac{3}{2}k_B$, though we will later see that it can be larger than this if the molecule can store energy in forms other than translational kinetic energy.⁷

The derivation of the thermal conductivity of a gas is very similar to that for viscosity. Consider molecules travelling along the z -axis. Then molecules which cross a plane of constant z will have travelled on average a distance λ since their last collision, and so they will have travelled a distance $\lambda \cos \theta$ parallel to the z -axis since their last collision. Therefore, they bring a deficit of thermal energy given by

$$C_{\text{molecule}} \times \Delta T = C_{\text{molecule}} \frac{\partial T}{\partial z} \lambda \cos \theta, \quad (9.16)$$

where C_{molecule} is the heat capacity of a single molecule. Hence the total thermal energy transported across unit area in unit time, i.e. the heat flux, is given by

$$\begin{aligned} J_z &= \int_0^\infty dv \int_0^\pi \left(-C_{\text{molecule}} \frac{\partial T}{\partial z} \lambda \cos \theta \right) v \cos \theta n f(v) \frac{1}{2} \sin \theta d\theta \\ &= -\frac{1}{2} n C_{\text{molecule}} \lambda \int_0^\infty v f(v) dv \frac{\partial T}{\partial z} \int_0^\pi \cos^2 \theta \sin \theta d\theta \\ &= -\frac{1}{3} n C_{\text{molecule}} \lambda \frac{\partial T}{\partial z}. \end{aligned} \quad (9.17)$$

⁴See Chapter 2.

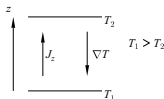


Fig. 9.7 Heat flows in the opposite direction to the temperature gradient.

⁵Thermal conductivity has units $\text{W m}^{-1} \text{K}^{-1}$.

⁶See Section 2.2.

⁷Other forms include rotational kinetic energy or vibrational energy, if the gas molecules are polyatomic.

Hence, the thermal conductivity κ is given by

$$\kappa = \frac{1}{3} C_V \lambda \langle v \rangle, \quad (9.18)$$

where $C_V = n C_{\text{molecule}}$ is the heat capacity per unit volume (though the subscript V here refers to a temperature change at constant volume). Equation 9.18 has some important consequences.

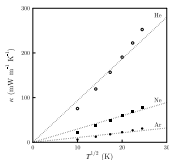


Fig. 9.8 The thermal conductivity of various gases as a function of temperature. The agreement with the predicted $T^{1/2}$ behaviour is satisfactory as a first approximation, but not very good in detail.

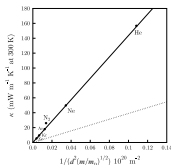


Fig. 9.9 The dependence of the thermal conductivity of various gases on $1/(\sqrt{md}^2)$. The dotted line is the prediction of eqn 9.19. The solid line is the prediction of eqn 9.46 which works very well for the monatomic noble gases, but a little less well for diatomic N_2 .

⁸See Section 2.2.

- κ independent of pressure.

The argument is the same as for η . Because $\kappa \approx 1/(\sqrt{2}n\sigma) \propto n^{-1}$, κ is independent of n and hence (at constant temperature) it is independent of pressure.

- $\kappa \propto T^{1/2}$.

The argument is also the same as for η . Because κ is independent of n , the only temperature dependence is from $\langle v \rangle \propto \sqrt{T}$, and hence $\eta \propto T^{1/2}$. This holds really quite well for a number of gases (see Fig. 9.8).

- As for viscosity, substituting in $\lambda = (\sqrt{2}n\sigma)^{-1}$, $\sigma = \pi d^2$ and $\langle v \rangle = (8k_B T/\pi m)^{1/2}$ yields a more useful (though less memorable) expression for the thermal conductivity:

$$\kappa = \frac{2}{3\pi d^2} C_{\text{molecule}} \left(\frac{k_B T}{\pi m} \right)^{1/2}. \quad (9.19)$$

- $L \gg \lambda \gg d$ is again the relevant condition for our treatment to hold.
- Equation 9.19 predicts that the thermal conductivity will be proportional to $1/(\sqrt{md}^2)$ at constant temperature. This holds very well, as shown in Fig. 9.9.
- Thermal conductivity can be measured by various techniques, see the box.

The similarity of η and κ would suggest that

$$\frac{\kappa}{\eta} = \frac{C_{\text{molecule}}}{m}. \quad (9.20)$$

The ratio C_{molecule}/m is the specific heat capacity⁸ c_V (the subscript V indicating a measurement at constant volume), so equivalently

$$\kappa = c_V \eta. \quad (9.21)$$

However, neither of these relations hold too well. Faster molecules cross a given plane more often than slow ones. These carry more kinetic energy and therefore do carry more heat. However, they don't necessarily carry more average momentum in the x -direction. We will return to this point in Section 9.4.

Measurement of thermal conductivity

The thermal conductivity κ can be measured using the *hot-wire method*. Gas fills the space between two coaxial cylinders (inner cylinder radius a , outer cylinder radius b) as shown in Fig. 9.10.

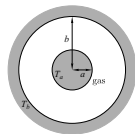


Fig. 9.10 The hot-wire method for measuring thermal conductivity.

The outer cylinder is connected to a constant-temperature bath of temperature T_b , while heat is generated in the inner cylinder (the hot wire) at rate Q per unit length of the cylinder (measured in units of W m^{-1}). The temperature of the inner cylinder rises to T_a . The rate Q can be connected with the radial heat flux J_r using

$$Q = 2\pi r J_r, \quad (9.22)$$

and J_r itself is given by $-\kappa \partial T / \partial r$, as in eqn 9.14. Hence

$$Q = -2\pi r \kappa \left(\frac{\partial T}{\partial r} \right), \quad (9.23)$$

and rearranging and integrating yields

$$Q \int_a^b \frac{dr}{r} = -2\pi \kappa \int_{T_a}^{T_b} dT, \quad (9.24)$$

and hence

$$\kappa = \frac{Q}{2\pi} \frac{\ln b/a}{T_a - T_b}. \quad (9.25)$$

Since Q is known (it is the power supplied to heat the inner cylinder) and T_a and T_b can be measured, the value of κ can be deduced.

An important application of this technique is in the **Pirani gauge**, which is commonly used in vacuum systems to measure pressure. A sensor wire is heated electrically, and the pressure of the gas is determined by measuring the current needed to keep the wire at a constant temperature. (The resistance of the wire is temperature dependent, so the temperature is estimated by measuring the resistance of the wire.) The Pirani gauge thus relies on the fact that at low pressure the thermal conductivity is a function of pressure (since the condition $\lambda \ll L$, where L is a linear dimension in the gauge, is not met). In fact, a typical Pirani gauge will not work to detect pressures much above 1 mbar because, above these pressures, the thermal conductivity of the gases no longer changes with pressure. The thermal conductivity of each gas is different, so the gauge has to be calibrated for the individual gas being measured.

9.3 Diffusion

Consider a distribution of similar molecules, some of which are labelled (e.g. by being radioactive). Let there be $n^*(z)$ of these labelled molecules per unit volume, but note that n^* is allowed to be a function of the z coordinate. The flux Φ_z of labelled molecules parallel to the z -direction (measured in $\text{m}^{-2}\text{s}^{-1}$) is⁹

$$\Phi_z = -D \left(\frac{\partial n^*}{\partial z} \right), \quad (9.26)$$

where D is the **coefficient of self-diffusion**.¹⁰ Now consider a thin slab of gas of thickness dz and area A , as shown in Fig. 9.11. The flux into the slab is

$$A \Phi_z, \quad (9.27)$$

⁹In three dimensions, this equation is written $\Phi = -D \nabla n^*$.

¹⁰We use the phrase self-diffusion because the molecules which are diffusing are the same (apart from being labelled) as the molecules into which they are diffusing. Below we will consider diffusion of molecules into dissimilar molecules.

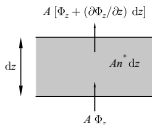


Fig. 9.11 The fluxes into and out of a thin slab of gas of thickness dz and area A .

¹¹See also Appendix C.12.

and the flux out of the slab is

$$A \left(\Phi_z + \frac{\partial \Phi_z}{\partial z} dz \right). \quad (9.28)$$

The difference in these two fluxes must be balanced by the time-dependent changes in the number of labelled particles inside the region. Hence

$$\frac{\partial}{\partial t} (n^* A dz) = -A \frac{\partial \Phi_z}{\partial z} dz, \quad (9.29)$$

so that

$$\frac{\partial n^*}{\partial t} = -\frac{\partial \Phi_z}{\partial z}, \quad (9.30)$$

and hence that

$$\frac{\partial n^*}{\partial t} = D \frac{\partial^2 n^*}{\partial z^2}. \quad (9.31)$$

This is the **diffusion equation**. A derivation of the diffusion equation in three dimensions is shown in the box.¹¹

Three-dimensional derivation of the diffusion equation

The total number of labelled particles that flow out of a closed surface S is given by the integral

$$\int_S \Phi \cdot d\mathbf{S}, \quad (9.32)$$

and this must be balanced by the rate of decrease of labelled particles inside the volume V which is surrounded by S , i.e.

$$\int_S \Phi \cdot d\mathbf{S} = -\frac{\partial}{\partial t} \int_V n^* dV. \quad (9.33)$$

The divergence theorem implies that

$$\int_S \Phi \cdot d\mathbf{S} = \int_V \nabla \cdot \Phi dV, \quad (9.34)$$

and hence that

$$\nabla \cdot \Phi = -\frac{\partial n^*}{\partial t}. \quad (9.35)$$

Substituting in $\Phi = -D \nabla n^*$ then yields the diffusion equation, which is

$$\frac{\partial n^*}{\partial t} = D \nabla^2 n^*. \quad (9.36)$$

A kinetic theory derivation of D proceeds as follows. The excess labelled molecules hitting unit area per second is

$$\begin{aligned} \Phi_z &= \int_0^\infty v \cos \theta f(v) dv \frac{1}{2} \sin \theta \left(-\frac{\partial n^*}{\partial z} \lambda \cos \theta \right) \\ &= -\frac{1}{3} \lambda(v) \frac{\partial n^*}{\partial z}, \end{aligned} \quad (9.37)$$

and hence

$$D = \frac{1}{3} \lambda \langle v \rangle. \quad (9.38)$$

This equation has some important implications:

- $D \propto p^{-1}$
In this case, there is no factor of n , but $\lambda \propto 1/n$ and hence $D \propto n^{-1}$ and at fixed temperature $D \propto p^{-1}$ (this holds quite well experimentally, see Fig. 9.12).
- $D \propto T^{3/2}$
Because $p = nk_B T$ and $\langle v \rangle \propto T^{1/2}$, we have that $D \propto T^{3/2}$ at fixed pressure.
- $D\rho = \eta$
The only difference between the formula for D and that for η is a factor of $\rho = nm$, and so

$$D\rho = \eta. \quad (9.39)$$

- $D \propto m^{-1/2} d^{-2}$ which is the same dependence as thermal conductivity.
- The less-memorable formula for D is, as before, obtained by substituting in the expressions for $\langle v \rangle$ and λ , yielding

$$D = \frac{2}{3\pi n d^2} \left(\frac{k_B T}{\pi m} \right)^{1/2}. \quad (9.40)$$

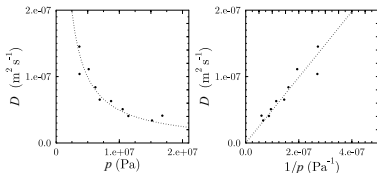


Fig. 9.12 Diffusion as a function of pressure.

This section has been about self-diffusion, where labelled atoms (or molecules) diffuse amongst unlabelled, but otherwise identical atoms (or molecules). Experimentally, it is easier to measure the diffusion of atoms (or molecules) of one type (call them type 1, mass m_1 , diameter d_1) amongst atoms (or molecules) of another type (call them type 2, mass m_2 , diameter d_2). In this case the diffusion constant D_{12} is used which is given by eqn 9.40 with d replaced by $(d_1 + d_2)/2$ and m replaced by $2m_1 m_2 / (m_1 + m_2)$, so that

$$D_{12} = \frac{2}{3\pi n \left(\frac{1}{2}(d_1 + d_2) \right)^2} \left(\frac{k_B T (m_1 + m_2)}{2\pi m_1 m_2} \right)^{1/2}. \quad (9.41)$$

9.4 More-detailed theory

The treatment of the transport properties presented so far in this chapter has the merit that it allows one to get the basic dependences fairly straightforwardly, and gives good insight as to what is going on. However, some of the details of the predictions are not in complete agreement with experiment and it is the purpose of this section to offer a critique of this approach and see how things might be improved. This section contains more advanced material than considered in the rest of this chapter and can be skipped at first reading.

One effect which we have ignored is the persistence of velocity after a collision. Our assumption has been that following a collision, a molecule's velocity becomes completely randomized and is completely uncorrelated with its velocity before the collision. However, although that is the simplest approximation to take, it is not correct. After most collisions, a molecule will retain some component of its velocity in the direction of its original motion. Moreover, our treatment has implicitly assumed a Maxwellian distribution of molecular velocities and that the different components of $\mathbf{v}$ are uncorrelated with each other, so that they can be considered to be independent random variables.¹² However, these components are actually partially correlated with each other and so are not independent random variables.

A further effect which becomes important at low pressure is the presence of boundaries; the details of the collisions of molecules with walls of a container can be quite important, and such collisions become more important as the pressure is reduced so that the mean free path increases.

Yet another consideration is the interconversion between the internal energy of a molecule and its translational degrees of freedom. As we will see in later chapters, the heat capacity of a molecule contains terms not only due to its translational motion ($C_{\text{molecule}} = \frac{3}{2}k_B$) but also due to its rotational and vibrational degrees of freedom. Collisions can give rise to processes where a molecule's energy can be redistributed throughout these different degrees of freedom. Thus if the molar heat capacity C_V can be written as the sum of two terms, $C_V = C'_V + C''_V$, where C'_V is due to translational degrees of freedom and C''_V is due to other degrees of freedom, then it turns out that eqn 9.21 should be amended to give

$$\kappa = \left(\frac{5}{2} C'_V + C''_V \right) \eta. \quad (9.42)$$

The $\frac{5}{2}$ factor reflects the correlations that exist between momentum, energy and translational motion. The most energetic molecules are the most rapid and therefore possess longer mean free paths. This leads to **Eucken's formula**, which states that

$$\kappa = \frac{1}{4}(9\gamma - 5)\eta C_V. \quad (9.43)$$

For an ideal monatomic gas $\gamma = \frac{5}{3}$ and hence

$$\kappa = \frac{5}{2}\eta C_V, \quad (9.44)$$

¹²See Section 3.6.

which supersedes eqn 9.21.

A more accurate treatment of the effects mentioned in this section has been performed by Chapman and Enskog (in the twentieth century); the methods used go beyond the scope of this text, but we summarize the results.

- The viscosity, which was written as $\eta = (2/3\pi d^2)(mk_{\text{B}}T/\pi)^{1/2}$ in eqn 9.6, should be replaced by

$$\eta = \frac{5}{16} \frac{1}{d^2} \left(\frac{mk_{\text{B}}T}{\pi} \right)^{1/2}, \quad (9.45)$$

i.e. the $2/3\pi$ should be replaced by $5/16$.

- The corrected formula for κ (which we had evaluated in eqn 9.19) can be obtained from this expression of η using Eucken's formula, eqn 9.43, and hence reads

$$\kappa = \frac{25}{32d^2} C_{\text{molecule}} \left(\frac{k_{\text{B}}T}{\pi m} \right)^{1/2}, \quad (9.46)$$

i.e. the $2/3\pi$ should be replaced by $25/32$.

- The formula for D , which appears in eqn 9.40, should now be replaced by

$$D = \frac{3}{8} \frac{1}{nd^2} \left(\frac{k_{\text{B}}T}{\pi m} \right)^{1/2}, \quad (9.47)$$

i.e. the $2/3\pi$ should be replaced by $3/8$. Similarly, eqn 9.41 should be replaced by

$$D = \frac{3}{8n(\frac{1}{2}[d_1 + d_2])^2} \left(\frac{k_{\text{B}}T(m_1 + m_2)}{2\pi m_1 m_2} \right)^{1/2}. \quad (9.48)$$

This also alters other conclusions, such as eqn 9.39, which becomes

$$D\rho = \frac{\frac{3}{8}\eta}{\frac{5}{16}} = \frac{6\eta}{5}. \quad (9.49)$$

Chapter summary

- Viscosity, η , defined by $\Pi_z = -\eta \partial \langle u_z \rangle / \partial z$ is (approximately)

$$\eta = \frac{1}{3} n m \lambda \langle v \rangle.$$

- Thermal conductivity, κ , defined by $J_z = -\kappa \partial T / \partial z$ is (approximately)

$$\kappa = \frac{1}{3} C_V \lambda \langle v \rangle.$$

- Diffusion, D , defined by $\Phi_z = -D \partial n^* / \partial z$ is (approximately)

$$D = \frac{1}{3} \lambda \langle v \rangle.$$

- These relationships assume that

$$L \gg \lambda \gg d.$$

The results of a more detailed theory have been summarized (which serve only to alter the numerical factors at the start of each equation).

- The predicted pressure, temperature, molecular mass and molecular diameter dependences are:

η	κ	D
$\propto p^0$	$\propto p^0$	$\propto p^{-1}$
$\propto T^{1/2}$	$\propto T^{1/2}$	$\propto T^{3/2}$
$\propto m^{1/2} d^{-2}$	$\propto m^{-1/2} d^{-2}$	$\propto m^{-1/2} d^{-2}$

(In this table, $\propto p^0$ means independent of pressure.)

Further reading

Chapman and Cowling (1970) is the classic treatise describing the more advanced treatment of transport properties in gases.

Exercises

- (9.1) Is air more viscous than water? Compare the dynamic viscosity η and the kinematic viscosity $\nu = \eta/\rho$ using the following data:

	ρ (kg m^{-3})	η (Pa s)
Air	1.3	17.4×10^{-6}
Water	1000	1.0×10^{-3}

- (9.2) Obtain an expression for the thermal conductivity of a gas at ordinary pressures. The thermal conductivity of argon (atomic weight 40) at S.T.P. is $1.6 \times 10^{-2} \text{ W m}^{-1} \text{ K}^{-1}$. Use this to calculate the mean free path in argon at S.T.P. Express the mean free path in terms of an effective atomic radius for collisions and find the value of this radius. Solid argon has a close-packed cubic structure in which, if the atoms are regarded as hard spheres, 0.74 of the volume of the structure is filled. The density of solid argon is $1.6 \times 10^3 \text{ kg m}^{-3}$. Compare the effective atomic radius obtained from this information with your effective collision radius. Comment on your result.

- (9.3) Define the coefficient of viscosity. Use kinetic theory to show that the coefficient of viscosity of a gas is given, with suitable approximations, by

$$\eta = K\rho\langle c \rangle \lambda$$

where ρ is the density of the gas, λ is the mean free path of the gas molecules, $\langle c \rangle$ is their mean speed, and K is a number which depends on the approximations you make.

In 1660 Boyle set up a pendulum inside a vessel which was attached to a pump which could remove air from the vessel. He was surprised to find that there was no observable change in the rate of damping of the swings of the pendulum when the pump was set going. Explain his observation in terms of the above formula.

Make a rough order of magnitude estimate of the lower limit to the pressure which Boyle obtained; use reasonable assumptions concerning the apparatus which Boyle might have used. [The viscosity of air at atmospheric pressure and at 293 K is $18.2 \mu\text{N s m}^{-2}$.]

Explain why the damping is nearly independent of pressure despite the fact that fewer molecules collide with the pendulum as the pressure is reduced.

- (9.4) Two plane disks, each of radius 5 cm, are mounted coaxially with their adjacent surfaces 1 mm apart. They are in a chamber containing Ar gas at S.T.P. (viscosity $2.1 \times 10^{-5} \text{ N s m}^{-2}$) and are free to rotate about their common axis. One of them rotates with an angular velocity of 10 rad s^{-1} . Find the torque which must be applied to the other to keep it stationary.

- (9.5) Measurements of the viscosity, η , of argon gas (^{40}Ar) over a range of pressures yield the following results at two temperatures:

$$\begin{array}{ll} \text{at } 500 \text{ K} & \eta \approx 3.5 \times 10^{-5} \text{ kg m}^{-1} \text{ s}^{-1}; \\ \text{at } 2000 \text{ K} & \eta \approx 8.0 \times 10^{-5} \text{ kg m}^{-1} \text{ s}^{-1}. \end{array}$$

The viscosity is found to be approximately independent of pressure. Discuss the extent to which these data are consistent with (i) simple kinetic theory, and (ii) the diameter of the argon atom (0.34 nm) deduced from the density of solid argon at low temperatures.

- (9.6) In Section 11.3, we will define the ratio of C_p to C_V is given by the number γ . We will also show that $C_p = C_V + R$, where the heat capacities here are per mole. Show that these definitions lead to

$$C_V = \frac{R}{(\gamma - 1)}. \quad (9.50)$$

Starting with the formulae $C_V = C'_V + C''_V$ and $\kappa = (\frac{5}{2}C'_V + C''_V)\eta$, show that if $C'_V/R = \frac{3}{2}$, then

$$\kappa = \frac{1}{4}(9\gamma - 5)\eta C_V, \quad (9.51)$$

which is Eucken's formula. Deduce the value of γ for each of the following monatomic gases measured at room temperatures.

Species	$\kappa/(\eta C_V)$
He	2.45
Ne	2.52
Ar	2.48
Kr	2.54
Xe	2.58

Deduce what proportion of the heat capacity of the molecules is associated with the translational degrees of freedom for these gases. (Hint: notice the word 'monatomic'.)